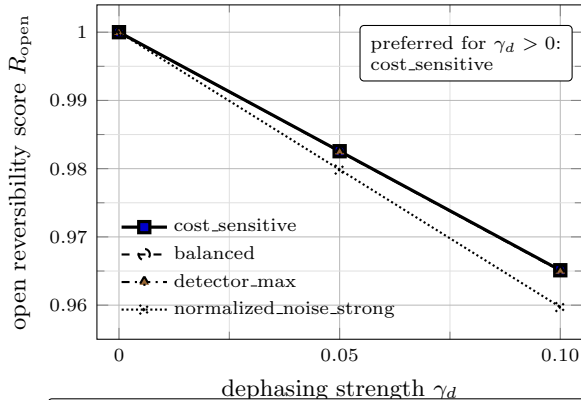
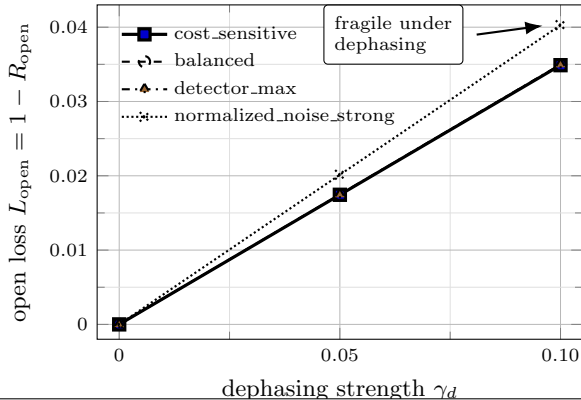


KTA Reversibility Ensemble: robustness under synthetic dephasing

(A) Open-system recoverability



(B) Dephasing-induced irreversibility



Audit reading. Forward propagation $U(t)$ followed by inverse propagation $U(-t)$ is coherently reversible to numerical precision.

Open-system loss appears only after a dephasing channel is inserted. Thus R_{open} and L_{open} are operating-mode metadata for hardware-native error suppression, not claims about time travel or fundamental time reversal.